AcousticDrivenGeneration

ETL report (processed data)

Source: C:/Users/dawoo/PycharmProjects/AcousticDrivenGeneration/processed

Tables: split sizes, simulated vs real counts, group counts (test),

overlap drops (example_hash), text length stats.

Figures: feature histograms and input/target length histograms per split.

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What this report contains

This document summarizes the output of the ETL pipeline: CSV splits under data/Data_splits/ are normalized, acoustic features are parsed from the Instructions field, and artifacts are written under processed/<train_size>/ (CSV/JSONL plus summary.json).

The tables quantify how many rows exist in each split, how many examples were removed to avoid exact duplicate leakage across splits, and how long the model input and target texts are. The figures show distributions of the seven parsed features and of character lengths for train, validation, and test.

This PDF includes the following simulated train sizes found on disk: 1k.

Table 1 — Dataset sizes and splits

Each row is one combination of train_size (which simulated training subset was used when building that processed folder) and split name.

The rows column is the number of examples kept after parsing all seven acoustic features and dropping rows with missing values. Simulated train/val rows have is_real False; the test split is real clinical data with is_real True.

group_counts is only meaningful for the real test split: PD vs HC counts. Simulated splits have no group label (shown as None).

Dataset sizes and splits

train_size	split	rows	is_real	group_counts
1k	train	1000	{"False": 1000}	{"None": 1000}
1k	val	10000	{"False": 10000}	{"None": 10000}
1k	test	96	{"True": 96}	{"HC": 48, "PD": 48}

Table 2 — Leakage / overlap handling

Simulated train and val CSVs reuse numeric IDs, so overlap by sample_id is not a reliable leakage check. The ETL pipeline deduplicates within each file and then removes val (and test) rows whose example_hash matches an example already present in train (or val, for test).

example_hash is SHA-256 of the normalized input_text plus the target report text, so it flags exact duplicate prompt+report pairs across splits.

val_dropped is the count of validation rows removed for matching train; test_dropped is the count of real test rows removed for matching train or val (usually zero). Small non-zero val_dropped values are expected when the same synthetic example appears in both train and val.

Leakage / overlap handling (rows removed from val/test if same example_hash as train)

train_size	val_dropped	test_dropped
1k	0	0

Table 3 — Text length statistics

input_text is the fixed-order string built from the seven parsed features (Breathing, Lips, Palate, Larynx, Monotonicity, Tongue, Intelligibility). target_text is the clinical report string.

Columns in_min / in_p50 / in_max are the minimum, median, and maximum length of input_text in characters for that split. tgt_* are the same for target_text.

Use this table to sanity-check tokenizer limits and to compare simulated vs real test report lengths.

Text length stats (characters)

train_size	split	in_min	in_p50	in_max	tgt_min	tgt_p50	tgt_max
1k	train	100	106	106	384	491	646
1k	val	98	106	106	384	494	646
1k	test	100	106	106	396	438	570

Figures — How to read the histograms

For each processed train size (1k, 10k, 100k when present), six images are included: train, val, and test, each with a features grid and a lengths pair.

Features plots: one histogram per parsed feature (breathing through intelligibility). They show the empirical distribution of values in that split after ETL.

Lengths plots: left histogram is input_text length in characters; right is target_text length. Together they show how compact prompts are relative to reports.

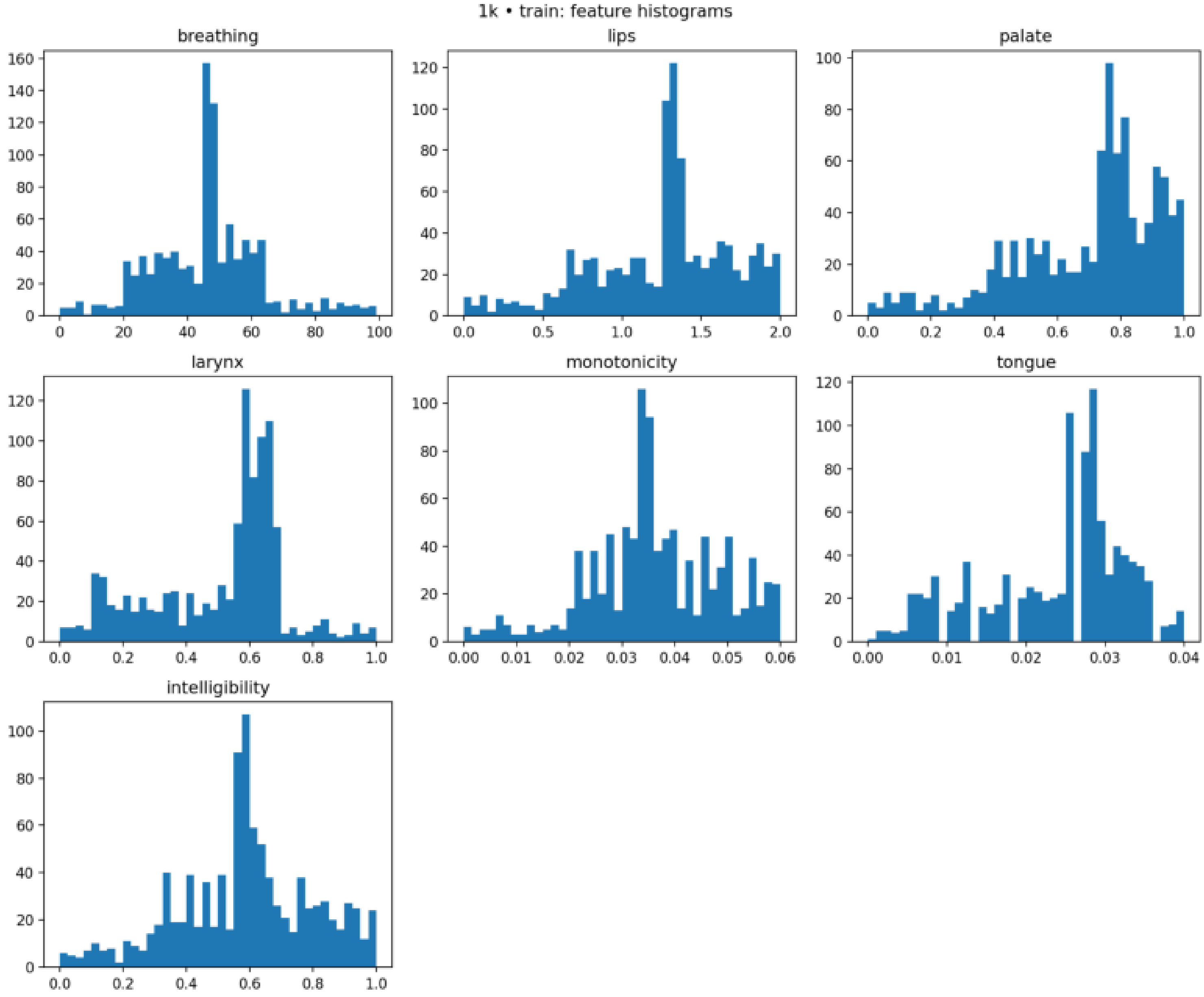
Simulated train/val distributions may differ slightly between train_size folders because the underlying sampled rows differ; the real test images are the same across folders (same 96 real examples), repeated for each train_size section.

Figures — train_size = 1k

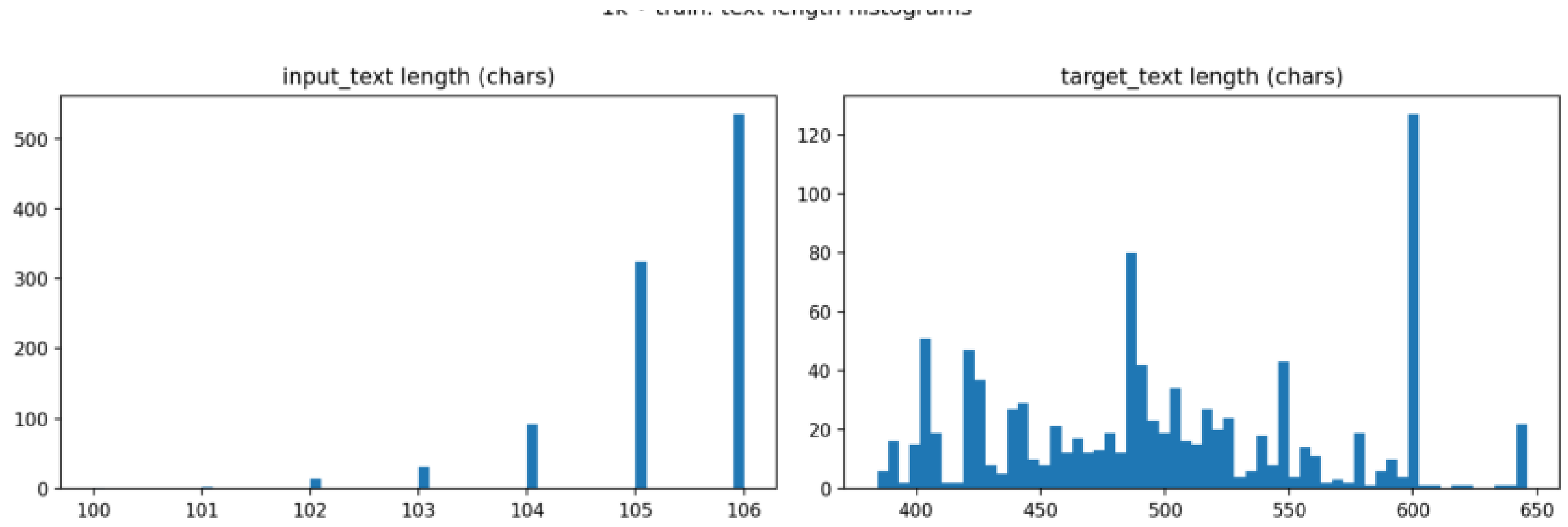
The following pages show ETL dashboard plots for the processed/1k folder: synthetic train and val (unless noted) and real test.

Order: train features, train lengths, val features, val lengths, test features, test lengths. If a PNG is missing, that page is skipped.

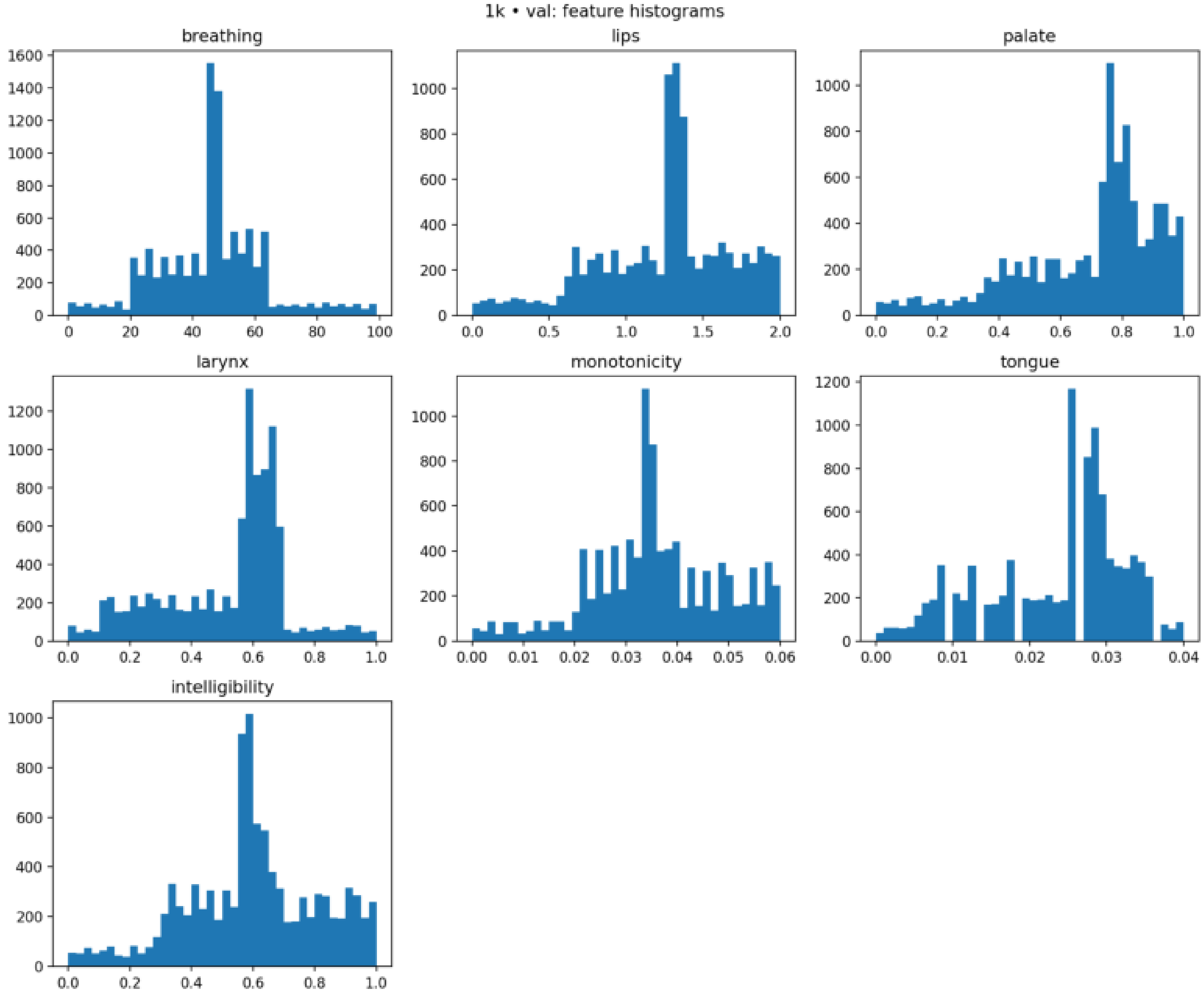
1k — train — features



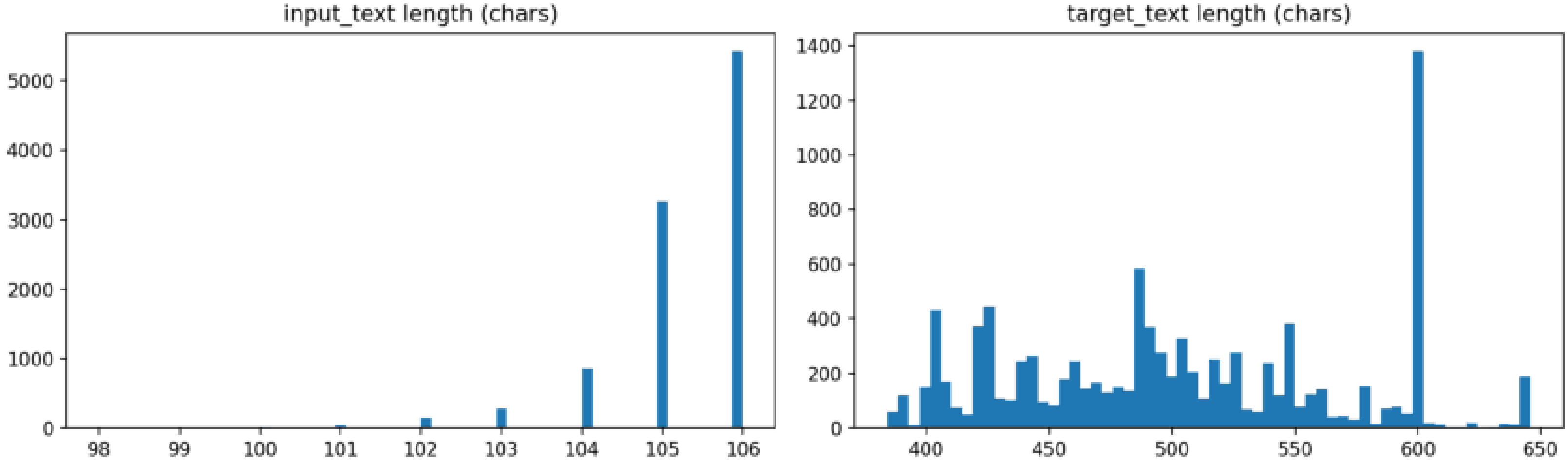
1k — train — lengths



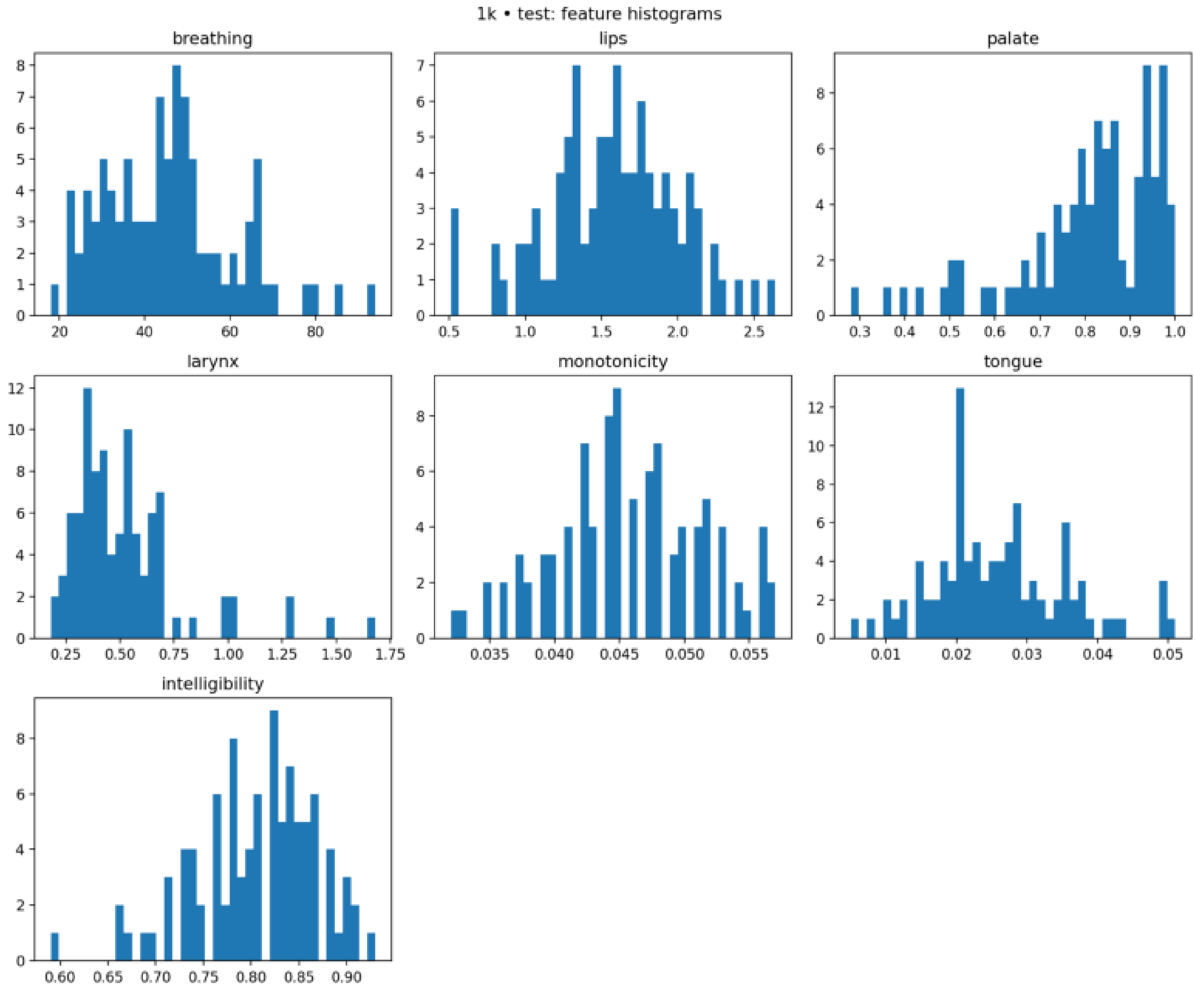
1k — val — features



1k — val — lengths



1k — test — features



1k — test — lengths